

Ex2-A114290023

A114290023

2026-09-30

```
print(uuid::UUIDgenerate())
```

```
## [1] "064f907e-c428-4414-be48-eb9b739bc5ee"
```

Data

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24,  
            31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23, 39)
```

Quation1

```
r <- max(commute) - min(commute)  
r      # range 27
```

```
## [1] 27
```

```
r / 5      # class width 5.4
```

```
## [1] 5.4
```

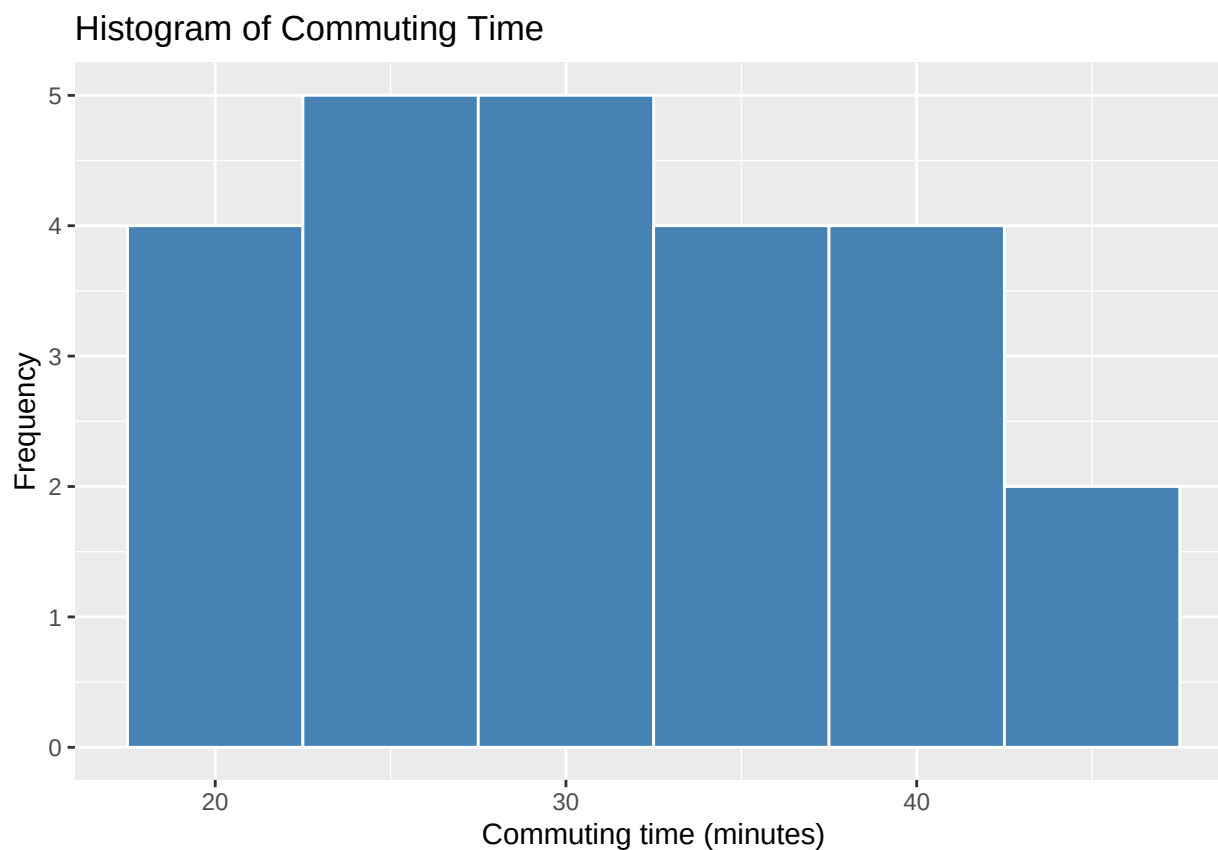
```
a <- cut(commute, 5)  
freq <- table(a)  
cum_freq <- cumsum(freq)  
cbind(Frequency = freq, Cumulative = cum_freq)
```

```
##           Frequency Cumulative  
## (18,23.4]         5           5  
## (23.4,28.8]       5          10  
## (28.8,34.2]       5          15  
## (34.2,39.6]       5          20  
## (39.6,45]        4          24
```

Quation2

```
library(ggplot2)
```

```
ggplot(data.frame(commute), aes(x = commute)) +  
  geom_histogram(binwidth = 5, fill = "steelblue", color = "white") +  
  labs(title = "Histogram of Commuting Time",  
        x = "Commuting time (minutes)", y = "Frequency")
```



Quation3

```
upper_boundaries <- c(23.4, 28.8, 34.2, 39.6, 45)  
ogive_data <- data.frame(upper_boundary = upper_boundaries,  
                          cumulative_frequency = as.numeric(cum_freq))  
  
ggplot(ogive_data, aes(x = upper_boundary, y = cumulative_frequency)) +  
  geom_line() +  
  geom_point() +  
  labs(title = "Ogive of Commuting Time",  
        x = "Upper class boundary (minutes)", y = "Cumulative frequency")
```

Ogive of Commuting Time

